NETWORK PROTOCOLS – QUESTIONS AND ANSWERS

Expanded Practical Assignment

TASK 1 — Capturing Spotify Network Traffic in Wireshark

Question 1: Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.

Answer

Wireshark is a network protocol analyser that records packets travelling through a selected network interface. It helps us inspect how a browser finds a website and establishes a secure connection. For this activity, DNS packets show how the domain name is resolved, while HTTPS-related packets show how the browser begins a protected connection.

Procedure

1. Open Wireshark and select the active Wi-Fi or Ethernet interface. The interface showing network activity is usually the correct one.
2. Start the capture before opening the website. In your browser, enter https://www.spotify.com and wait for the page to load.
3. Return to Wireshark and stop the capture. Enter the display filter dns to show DNS traffic.
4. Identify a DNS Standard query and its matching Standard query response. Record the domain name, source, destination and packet number.
5. Next, use the display filter tls to inspect TLS packets. Look for a Client Hello and Server Hello, which are part of the TLS handshake.

Wireshark — Illustrative packet capture

Display filter: dns || tls

No.	Time	Source	Destination	Protocol	Info
18	1.204	192.168.1.20	192.168.1.1	DNS	Standard query A www.spotify.com
19	1.231	192.168.1.1	192.168.1.20	DNS	Standard query response A ...
23	1.510	192.168.1.20	203.0.113.10	TLSv1.3	Client Hello
24	1.548	203.0.113.10	192.168.1.20	TLSv1.3	Server Hello

Example addresses and packet numbers are fictional; your capture will differ.

Figure 1. Illustrative Wireshark-style capture showing DNS query/response and TLS handshake packets. This is a teaching example, not a capture from your computer.

Sample packet observations

Packet type	Example Info field	Purpose	What to record
DNS query	Standard query A www.spotify.com	Requests the website's IPv4 address	Actual packet no., source and destination
DNS response	Standard query response A	Returns the address or	Actual answer and

	…	addresses	response code
TLS Client Hello	Client Hello	Browser begins TLS negotiation	Actual packet no. and server IP
TLS Server Hello	Server Hello	Server responds during TLS negotiation	Actual packet no. and protocol version

Important: Modern Spotify connections use HTTPS. You may see TLS or QUIC packets rather than readable HTTP messages. A TLS handshake is evidence of secure-connection setup, but it is not the same as seeing the encrypted web-page contents.

TASK 1 — Results, Explanation and Conclusion

How to interpret the capture

A DNS query is normally sent by the computer to a DNS resolver. The resolver replies with the address information needed to contact the requested service. After name resolution, the browser can establish a connection to the server. With HTTPS, TLS is used to negotiate encryption and protect data in transit.

Example observation table

Observation	Illustrative result	Meaning
DNS packet 1	Standard query for www.spotify.com	The computer is asking for the domain's address.
DNS packet 2	Standard query response	The resolver is returning address information.
HTTPS packet 1	TLS Client Hello	The browser proposes TLS settings.
HTTPS packet 2	TLS Server Hello	The server selects TLS settings and continues the handshake.

Why actual packet details may differ

- The DNS answer may already be cached, so a fresh DNS query may not appear in every capture.
- The website may use multiple IP addresses or content-delivery servers.
- The browser and server may use TLS 1.3 or QUIC, depending on the connection.
- Packet numbers, timestamps and source/destination addresses depend on the computer and network.

Conclusion

This activity demonstrates that opening a website involves more than sending a simple web request. DNS first helps locate the service, and TLS helps establish a secure connection for HTTPS communication. Wireshark makes these steps visible as individual packets.

Evidence to add from your own computer

For a fully verified submission, replace or supplement Figure 1 with your own Wireshark screenshot. The screenshot should show the dns or tls display filter and the relevant packet rows. Do not copy the example packet numbers or addresses as if they were measured on your device.

TASK 2 — DNS Lookup Using nslookup

Question 2: Simulate a DNS lookup for www.flipkart.com using the nslookup command in your terminal and write down the IP address returned along with a brief explanation of what this result means.

Answer

The nslookup command is a network utility used to query the Domain Name System. It can show which DNS server answered the request and the IP address associated with a domain name.

Procedure

6. On Windows, open Command Prompt. On macOS or Linux, open Terminal.
7. Type nslookup www.flipkart.com and press Enter.
8. Wait for the DNS response. Note the DNS server name/address and all IP addresses shown under the answer.
9. If more than one address is returned, record each address. Different resolvers or locations may return different results.

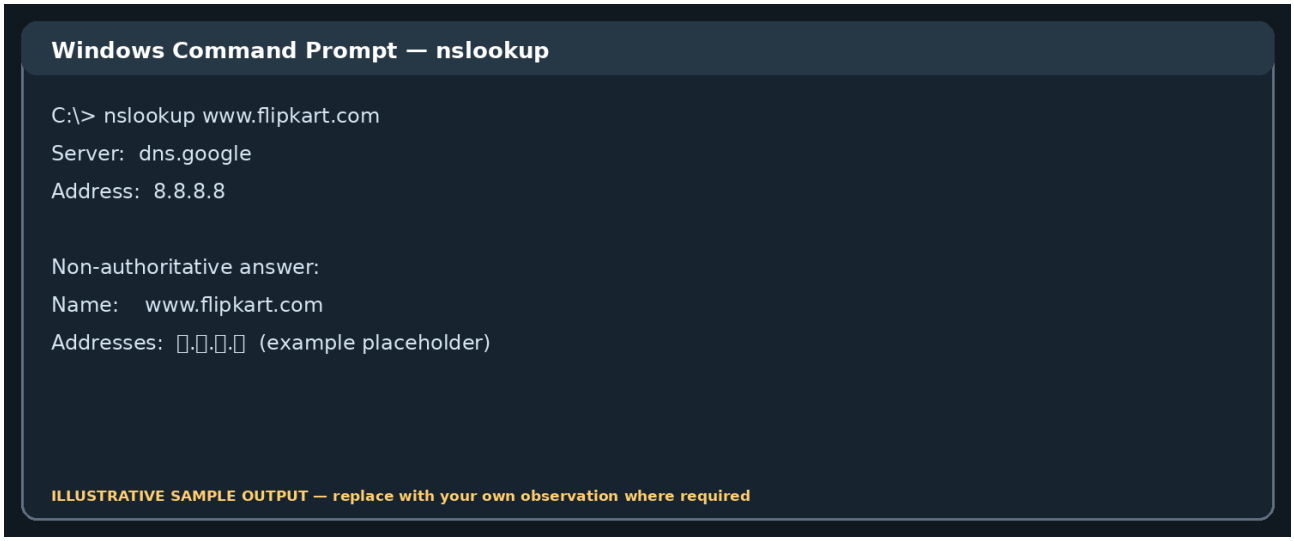


Figure 2. Example nslookup terminal display. The address shown is a placeholder; use the actual output from your own command.

Result and explanation

Observed IP address: _____

DNS server used: _____

The result means that the DNS resolver translated the human-readable name www.flipkart.com into one or more numerical IP addresses. Computers use IP addresses to route network traffic to the correct destination. DNS works like a directory: a user enters a memorable website name, and DNS helps find the address needed to reach it.

Additional points

- An IPv4 address is written as four numbers separated by dots, for example 192.0.2.10 (documentation example only).
- An IPv6 address is longer and uses hexadecimal groups separated by colons.
- A DNS answer can change due to load balancing, content delivery networks, server changes or geographic location.

Conclusion

The nslookup command provides a quick way to test DNS name resolution and troubleshoot cases where a website name cannot be translated into an IP address.

TASK 3 — Difference Between HTTP and HTTPS

Question 3: Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.

Answer

HTTP (Hypertext Transfer Protocol) is used to transfer web resources between a browser and a web server. HTTPS is HTTP communication protected by TLS (Transport Layer Security). HTTPS is commonly used for websites that handle accounts, payments, personal details or other information that should be protected.

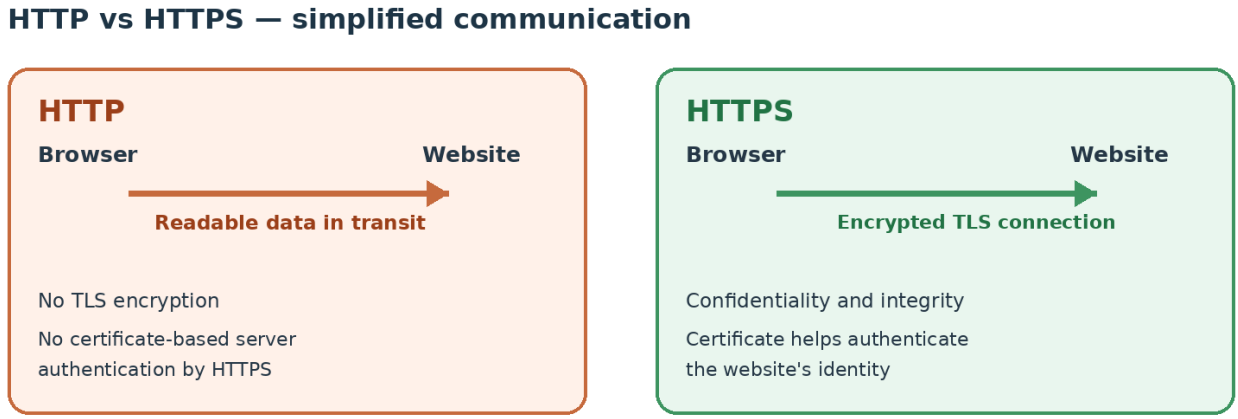


Figure 3. Simplified comparison of unencrypted HTTP and TLS-protected HTTPS communication.

Three key differences

Feature	HTTP	HTTPS
Encryption	Does not provide TLS encryption by itself.	Uses TLS to encrypt data in transit.
Authentication	Does not provide HTTPS certificate-based server authentication.	Uses a digital certificate to help authenticate the website.
Integrity	Does not provide TLS protection against undetected alteration in transit.	TLS helps detect tampering and protects data integrity.

Real-world example: Spotify

When a user opens Spotify's website using an HTTPS address, the browser establishes a TLS-protected connection. This helps protect information exchanged with the service, including account-related information, while it travels across the network. HTTPS does not guarantee that a website is trustworthy in every respect, but it provides important protection for the connection.

Why HTTPS matters

- It reduces the risk of other people on the same network reading the contents of the communication.
- It helps prevent data from being silently modified while travelling between the browser and server.
- It helps the browser verify that it is communicating with the site identified by the certificate.

Conclusion

HTTPS is preferred for modern websites because it adds encryption, integrity protection and certificate-based authentication to ordinary HTTP communication.

TASK 4 — Observing DNS Queries and Responses in Wireshark

Question 4: Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe. Hint: Look for packets with 'Standard query' and 'Standard query response' in the Info column.

Answer

DNS resolution is the process of finding the IP address associated with a domain name. A browser needs this information before it can normally connect to the website server.

Procedure

10. Open Wireshark, select the active interface and start capturing.
11. Enter dns in the display-filter bar. The filter displays packets identified as DNS traffic.
12. Visit a domain that has not been accessed recently, or use a test domain approved by your instructor.
13. Stop the capture and inspect the Info column for Standard query and Standard query response entries.
14. Match the query and response using the DNS transaction ID, requested name and record type.

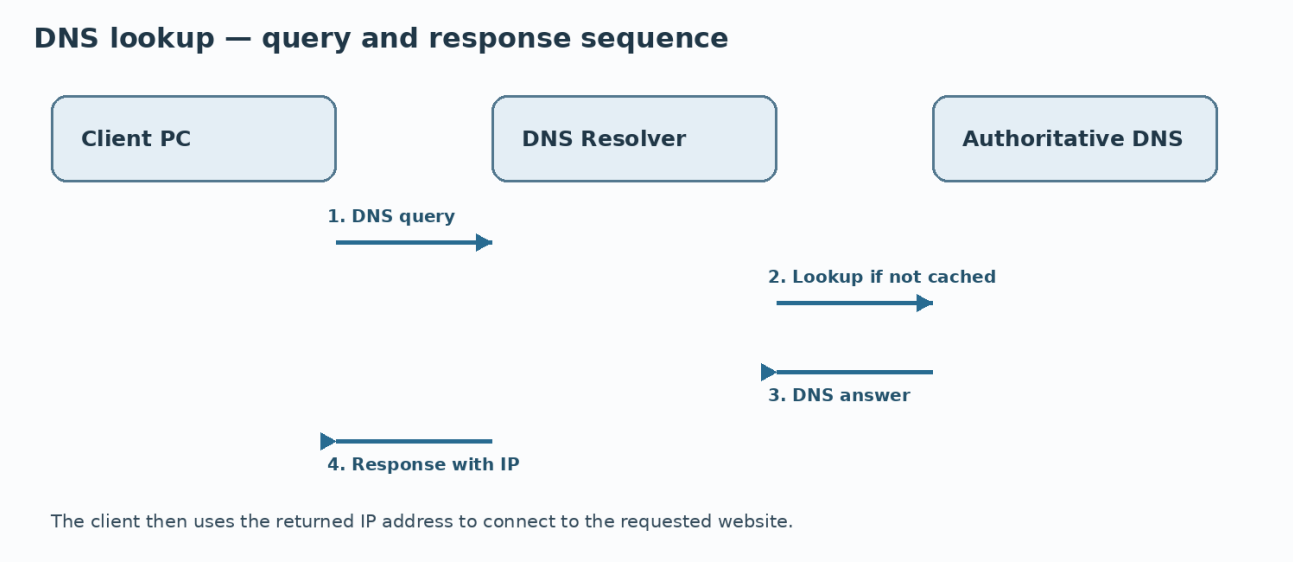


Figure 4. Simplified DNS resolution path. The resolver may use cached information or contact other DNS servers.

Typical sequence observed

15. The client sends a DNS query, often requesting an A record for an IPv4 address and/or an AAAA record for an IPv6 address.
16. The DNS resolver checks its cache. If the answer is not cached, it may contact other DNS servers to obtain the information.
17. The resolver sends a DNS response containing the requested address information, or an error such as a name-not-found response.
18. The computer uses the returned IP address to begin connecting to the website.

What to record

Field	Query packet	Response packet
Info	Standard query A/AAAA [domain]	Standard query response
Transaction ID	Record the ID shown	Should match the corresponding query

Requested name	Website domain	Same domain in the answer
Answer	Usually not present in query	Returned IP address or error

Conclusion

The capture shows the request-and-response pattern used by DNS. The exact sequence can vary because of caching, IPv4/IPv6 support and the configuration of the network's DNS resolver.

TASK 5 — Real-World Use of DHCP

Question 5: Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to Wi-Fi when you open Instagram at a cafe).

Answer

Selected protocol: DHCP (Dynamic Host Configuration Protocol)

Scenario: When I connect my phone to the Wi-Fi network at a cafe so that I can use Instagram, the phone needs network settings before it can communicate over the Internet. DHCP automatically provides the phone with network configuration, so I do not have to enter every setting manually.

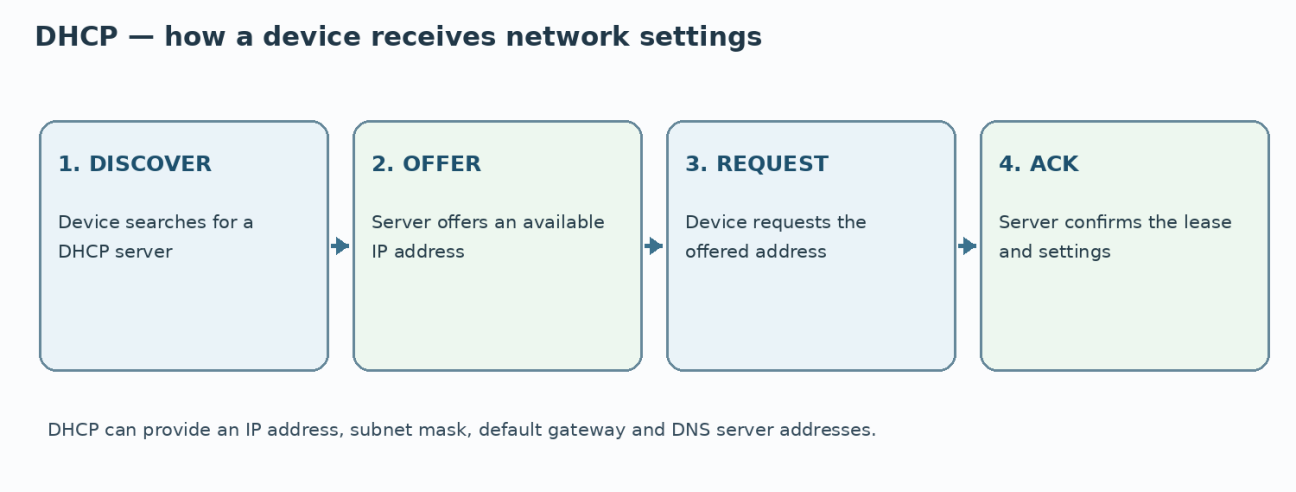


Figure 5. Simplified DHCP address assignment sequence: Discover, Offer, Request and Acknowledgement.

How DHCP works

19. DHCP Discover: The phone broadcasts a message to locate an available DHCP server on the local network.
20. DHCP Offer: The server offers an available IP address and configuration details.
21. DHCP Request: The phone requests the offered address.
22. DHCP Acknowledgement: The server confirms the address lease and provides settings such as subnet mask, default gateway and DNS server addresses.

Why DHCP is useful

- It reduces the need to configure IP settings manually on every device.
- It helps prevent address conflicts by managing a pool of available IP addresses.
- It can provide a lease for a limited period, allowing addresses to be reused when devices disconnect.

Conclusion

DHCP makes connecting to Wi-Fi networks convenient. Once the phone receives its IP address and other network settings, it can communicate with the local network and reach Internet services such as Instagram.

Final assignment conclusion

These five activities demonstrate important roles played by common network protocols. Wireshark helps observe packets, DNS translates domain names into IP addresses, HTTPS protects web communication, DNS query/response packets show the name-resolution process, and DHCP automatically configures devices when they join a network.

Screenshot checklist for final submission

- Task 1: Your own Wireshark capture showing at least two DNS packets and two TLS/HTTPS-related packets.
- Task 2: Your own Command Prompt or Terminal output for nslookup www.flipkart.com.
- Task 4: Your own Wireshark capture filtered with dns showing a query and its matching response.

The diagrams and sample terminal/capture visuals in this document are illustrative learning aids. They are not evidence that commands were run or packets were captured on your computer. Add your own screenshots wherever the instructor expects practical proof.